



Tuning the LAMP Stack to Boost the Performance of Drupal

The previous article in this series dealt with the various external factors that impact the performance of Drupal sites, exploring ways to mitigate them. Now, let's look at how it's possible to increase performance without additional hardware, but by tuning the various components of the hosting stack.

Early optimisation is bad and ought to be avoided. You should embark on an optimisation exercise only when you encounter issues like degradation in response times. First analyse the reasons for the slow response of each component of the technology stack before attempting a tuning exercise. But before going further, let's assume that your site is hosted on the LAMP stack, i.e., Linux (Centos or Ubuntu), Apache2, MySQL and PHP with Drupal 7.x; and that you have your site hosted on a VPS like Rackspace, Instacompute or on a dedicated server but not on a shared hosting environment. Let's also assume that you are able to log in to the VPS using SSH to gain access to the shell.

Understanding your stack's performance bottlenecks

Studies by SourceLabs have revealed that Apache is bandwidth limited, PHP is CPU limited and MySQL is memory limited. In our exercise, let's ensure that these components use what's available effectively rather than throwing in more of these resources.

Tools for measuring performance and monitoring

Firebug and Yslow

The Yslow add-on for Firebug (on Firefox, Chrome users have to use Firebug Light) is provided courtesy of the Yahoo Developer Network and provides a performance grade for sites along with an overall score.

Pingdom and Site24x7

The simplest way to measure performance is to use the several online tools available from Pingdom and Zoho. Pingdom is similar to Yslow; however, unlike the latter, which only tests performance from your browser, Pingdom allows testing of the site from several locations. This gives you an insight into the loading times in a particular region. In the rest of this article, I will use Pingdom to measure response times. Site24x7 is a similar offering from Zoho.

Munin

Munin is an invaluable network resource monitoring tool. A default installation provides information on server health in graphs. You can monitor the performance of almost any